

Incident Report — Cloud Identity Under Siege

SOC Analysis & Triage · Intermediate Operation

1. Header / Metadata

Operation ID	OPS-2025-1006-09
Operation name	Cloud Identity Under Siege (Cloud Token Theft)
Started (UTC)	2025-10-06 08:15:05
Closed (UTC)	2025-10-14 04:59:52
Analyst	Ndumiso Mbanjwa
Cohort	May26 Cybersecurity
Difficulty	Intermediate
Severity	Critical
Final verdict	True Positive
Status	Closed – Escalated for identity, Kubernetes-secret and storage-key rotation; perimeter egress controls to be reviewed

2. Executive Summary

This case covered nine days of activity (5–14 October 2025) across a cloud-first Azure company, “acme-corp”: Entra ID (Azure AD) sign-ins, Windows computers, a Kubernetes container cluster, an Azure Function and the storage accounts behind them. There was no ready-made alert pointing at the problem — the job was to notice what did not fit the normal pattern and rebuild the story from the records. The pattern that emerged centres on the workstation CORP-WKS-8821 (internal address 10.0.1.134), used by marcus.aurelius.

First, I confirmed what a normal, successful Windows logon looks like (Event ID 4624), so that later sign-ins could be judged against it. Then the anomalies stacked up. An unsigned program called goblob.exe, run from Marcus's Downloads folder, looked up the company's “finance-backups” cloud storage address. PowerShell on the same machine used the AADInternals hacking toolkit to request a Microsoft Graph access token — a digital key that lets a program act as a user without a password prompt — and that token was then used from 185.220.101[.]35, a Tor anonymity-network address, with no MFA. AzureHound, a tool attackers use to map every user, group and permission in a company's Azure directory, ran under the dev_ops_automation service account and saved its results

to /tmp/graph_objects.json. The workstation opened an unusual inbound connection to the production app server srv-app-prod-01, the emergency “break-glass” administrator account signed in from another internal machine (10.0.2.69) without MFA, kubectl.exe (the Kubernetes command-line tool) was used to dump every secret in the cluster to /tmp/etcd_secrets.txt, and a .env configuration file — the kind that usually holds passwords and connection strings — was pulled off srv-app-prod-01 over plain HTTP. Finally, 95,000 objects (4.2 GB) were read from the finance-backups storage container from the same Tor address.

The single most important operational finding is that the perimeter firewall stopped none of it: the outbound connection made when goblob.exe reached out was recorded with the verdict ALLOWED, as were the later connections to the attacker's addresses. Detection, not prevention, was the only control that worked. This is a confirmed True Positive: I resolved all 8 of the platform's scored tasks, and the verdict rests on the SIEM, XDR, firewall and cloud records and the platform's own ATT&CK Cloud mapping for this incident.

3. Operation Details

Triggering detection	None seeded — found by hunting. The investigation was opened to find what “does not fit the shape the estate normally has” across nine days of telemetry.
Source	CORP-WKS-8821 (10.0.1.134, marcus.aurelius and the dev_ops_automation service account); internal host 10.0.2.69 (break-glass sign-in); external 185.220.101[.]35 (Tor exit, token use and storage read) and 217.44.76[.]238
Destination	Entra ID / Microsoft Graph; production app server srv-app-prod-01; Kubernetes API server 10.0.2.143:6443; finance-backups blob storage container
User / Host affected	marcus.aurelius, dev_ops_automation, admin.breakglass; CORP-WKS-8821, srv-app-prod-01, k8s-node-primary-01
Artefact	goblob.exe; AADInternals Get-AADIntAccessToken; AzureHound.exe → /tmp/graph_objects.json; kubectl get secrets → /tmp/etcd_secrets.txt; .env copied to C:\Users\marcus.aurelius\AppData\Local\Temp\
Direction	Internal: workstation → app server, Kubernetes API and break-glass sign-in. Outbound: DNS/HTTPS to 217.44.76[.]238 and 185.220.101[.]35 (all ALLOWED). Cloud: storage read from Tor
Tool surfaces used	Security log platform (SIEM), endpoint detection (XDR), firewall (FW / Network) and cloud telemetry (CLOUD)

4. Investigation & Triage

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
Task 1 – Baseline for successful authentication	Learn what a normal successful logon looks like before judging anomalies	Windows security logs for marcus.aurelius's early sign-ins on corp-wks-8821	2025-10-06 08:59:53 and 14:02:11: “[4624] Successful logon for marcus.aurelius” on corp-wks-8821	Event ID 4624 is the Windows record of a successful logon. Every later sign-in carries it, so it is the yardstick for spotting unusual accounts, hosts and times	Fig. 2
Task 2 – Unexpected inbound connection to the app server	Find the internal source of a flow that does not match normal traffic	The CLOUD panel: NSG (network security group) flow records	Of three NetworkFlowRecorded events, only 2025-10-08 22:02:53 is flagged Suspicious: source 10.0.1.134, affected host srv-app-prod-01, result Success, mapped to T1021 Remote Services	The source is 10.0.1.134 — Marcus's workstation, not a server that normally talks to production. A user laptop reaching straight into the production app server is a lateral-movement signal	Fig. 3
Task 3 – Internal origin of the break-glass sign-in	Find where the emergency admin account signed in from	Azure AD sign-in records (CLOUD panel) filtered for breakglass	2025-10-09 10:59:53 UserLoggedIn for admin.breakglass, IP 10.0.2.69, MFA Not used, result Success, application srv-app-prod-01 (Suspicious). The account's other sign-ins came from 10.0.1.123	Break-glass accounts should almost never be used. A successful, MFA-less sign-in from a different internal host, 10.0.2.69, against production is a privileged-access red flag	Fig. 4
Task 4 – Client used against the	Name the non-control-plane	SIEM process and network-connection events for k8s-	Sysmon event: “Network connection detected:	kubelet.exe is the node's own agent; kubectl.exe is an administrator's	Fig. 5

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
Kubernetes API	process talking to the API server on port 6443	node-primary-01	kubectl.exe -> 10.0.2.143:6443"; kubelet.exe also connects to the same endpoint as normal node software	command-line client. Its presence on the node is what the XDR detections for container commands and secret listing (T1609, T1552.007) point to	
Task 5 – Staged directory-enumeration output	Recover where the Graph directory dump was written	Process-creation telemetry (Event 4688) and the XDR process tree on corp-wks-8821	2025-10-08 01:37:19, user dev_ops_automation, parent powershell.exe: "cmd.exe /c AzureHound.exe -u dev_ops_automation@acme-corp.cloud --collect all > /tmp/graph_objects.json"; graph_objects.json flagged Suspicious	The output path is /tmp/graph_objects.json. AzureHound maps every identity and permission in Azure — a blueprint for privilege escalation — and was run under a service account to blend in	Fig. 6, 7
Task 6 – Staged Kubernetes secrets file	Recover where the cluster secrets were written	Process-creation telemetry for marcus.aurelius on corp-wks-8821	2025-10-10 20:46:21 Event 4688: "cmd.exe /c kubectl get secrets --all-namespaces -o yaml > /tmp/etcd_secrets.txt"; matching XDR detection T1552.007 "kubectl.exe listed secrets across all namespaces"	The path is /tmp/etcd_secrets.txt. Dumping every namespace's secrets gives the attacker the passwords, keys and tokens the cluster's applications use	Fig. 8
Task 7 – Configuration file copied off	Name the file pulled from srv-	Process-creation and file-creation	2025-10-12 22:10:51 Event 4688 (parent powershell.exe,	The artefact is .env. These files hold application secrets such as database	Fig. 9

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
the app server	app-prod-01	telemetry on corp-wks-8821	user marcus.aurelius): "cmd.exe /c curl -s http://srv-app-prod-01/.env -o C:\Users\marcus.aurelius\AppData\Local\Temp\.env"; .env flagged Suspicious	passwords and API keys, and it was served over unencrypted HTTP from production	
Task 8 – Perimeter verdict on the outbound connection	Establish whether the firewall blocked anything	The firewall traffic log filtered for 10.0.1.134	2025-10-06 08:15:03: 10.0.1.134:49213 → 217.44.76.238:53 UDP, rule Allow-Outbound-DNS-Any, process goblob.exe, verdict Allowed. Later 10.0.1.134 → 185.220.101.35:443 and → 217.44.76.238:443 (powershell.exe) are also Allowed	The verdict is ALLOWED. This is the most operationally important finding: the perimeter prevented nothing, and every stage was caught only by detection and investigation	Fig. 10

5. Threat Intelligence

Indicator (defanged)	Tool	Result / score	Date checked (UTC)	Notes
185.220.101[.]35	VirusTotal	17 out of 89 vendors flagged it (malicious/malware/phishing); tagged "tor"; AS 60729 (Stiftung Erneuerbare Freiheit), DE;	2026-09-15	Strong outside confirmation that this is a Tor exit node. Legitimate staff should never reach Microsoft Graph or company storage through Tor, which matches the XDR token-reuse and 4.2

Indicator (defanged)	Tool	Result / score	Date checked (UTC)	Notes
		community score –9		GB storage-read detections.
217.44.76[.]238	VirusTotal	0 out of 89 vendors (unrated); AS 2856 (British Telecommunications), GB	2026-09-15	A clean score is not proof of safety. Our firewall shows goblob.exe using it for DNS and PowerShell repeatedly connecting to it on port 443 — unusual for a residential/ISP address.
goblob.exe (SHA-256 7b1f3c9a...a25e91)	VirusTotal	Never seen by VirusTotal (“Item not found”)	2026-09-15	Only means the file was never submitted. It is unsigned, ran from Downloads and is flagged Suspicious by the XDR.
AzureHound.exe (SHA-256 c4e9a07f...1d7e82)	VirusTotal	Never seen by VirusTotal (“Item not found”)	2026-09-15	This exact build was never submitted; AzureHound itself is a well-known Azure reconnaissance tool.
10.0.1.134 and 10.0.2.69	VirusTotal	Not applicable — private (internal) addresses	2026-09-15	Internal RFC 1918 addresses have no public reputation. They are indicators only inside this estate.

This is assessed with high confidence as a cloud identity and token-theft intrusion: an attacker operating from a user workstation stole a Microsoft Graph token with AADInternals, mapped the Azure directory with AzureHound, reached into production and Kubernetes, harvested secrets and configuration files, and read 4.2 GB from finance backup storage through Tor. Outside intelligence only strongly confirms one indicator — the Tor exit 185.220.101[.]35 (17 of 89). The other external address is unrated and both tool hashes are unknown to VirusTotal, while the internal addresses have no public reputation at all. A clean or missing VirusTotal result is treated as inconclusive, not as proof of safety; the findings rest on the organisation's own SIEM, XDR, firewall and cloud records. The dataset also contains deliberate decoys (for example a routine svc_az_backup storage job and an EICAR test file), which were checked and excluded.

6. Timeline (UTC)

Time (UTC)	Event
2025-10-06 08:15:03–08:15:05	goblob.exe (unsigned, Downloads) on CORP-WKS-8821 looks up finance-backups.blob.core.windows.net via 217.44.76[.]238:53 — firewall verdict ALLOWED
2025-10-06 08:59:53	Baseline: Event 4624 successful logon for marcus.aurelius on corp-wks-8821
2025-10-07 10:46:01	Critical — PowerShell runs Get-AADIntAccessToken for Microsoft Graph (T1528)
2025-10-08 01:37:19	High — AzureHound.exe under dev_ops_automation enumerates the directory and writes /tmp/graph_objects.json (T1087.004)
2025-10-08 02:09:53	Critical — Graph sign-in for marcus.aurelius from Tor exit 185.220.101[.]35 with no MFA, 9 minutes after a sign-in from 23.242.239.65 (T1550.001); firewall ALLOWED
2025-10-08 22:02:53	High — unexpected inbound NSG flow from 10.0.1.134 to srv-app-prod-01 (T1021)
2025-10-09 10:59:53	High — admin.breakglass signs in from 10.0.2.69 to srv-app-prod-01 without MFA
2025-10-10 09:29:56	High — AddAppRoleAssignment by marcus.aurelius from 10.0.1.134
2025-10-10 20:46:21	High — kubectl get secrets --all-namespaces written to /tmp/etcd_secrets.txt (T1552.007)
2025-10-12 22:10:51	High — .env pulled from http://srv-app-prod-01/.env into marcus.aurelius's Temp folder
2025-10-14 04:30:05	kubectl.exe connection to the Kubernetes API 10.0.2.143:6443 recorded on k8s-node-primary-01
2025-10-14 04:59:52	Critical — 95,000 objects (4.2 GB) read from the finance-backups container by dev_ops_automation from 185.220.101[.]35 (T1537)

7. Indicators of Compromise

Type	Indicator (defanged)	Context
IPv4 (internal)	10.0.1.134	CORP-WKS-8821; source of the unexpected inbound flow to srv-app-prod-01 — platform Confirmed finding
IPv4 (internal)	10.0.2.69	Origin of the MFA-less admin.breakglass sign-in — platform Confirmed finding
IPv4	185.220.101[.]35	Tor exit used for Graph token replay and the 4.2 GB finance-backups read

Type	Indicator (defanged)	Context
IPv4	217.44.76[.]238	External address used by goblob.exe (DNS) and PowerShell (HTTPS) — all ALLOWED
Process	kubectrl.exe	Non-control-plane client against the Kubernetes API 10.0.2.143:6443 — platform Confirmed finding
File path	/tmp/graph_objects.json	AzureHound directory-enumeration output — platform Confirmed finding
File path	/tmp/etcd_secrets.txt	Dump of all Kubernetes secrets — platform Confirmed finding
File name	.env	Configuration/secrets file copied from srv-app-prod-01 to the user's Temp folder — platform Confirmed finding
SHA-256	7b1f3c9a2e8d4f60a15c83be6d9072f4ab3e1c5d8f902a47b6c1de03f8a25e91	goblob.exe (unsigned, Suspicious)
SHA-256	c4e9a07f1d3b6582ae40f9c7b2d815630e9a4c1f78b50d2369ef0a4c5b1d7e82	AzureHound.exe (Suspicious)

STIX 2.1 Bundle

```
{
  "type": "bundle",
  "id": "bundle--25d17863-b563-592b-97ca-0450d2c4b9d0",
  "objects": [
    {
      "type": "identity",
      "spec_version": "2.1",
      "id": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
      "created": "2025-10-06T08:15:05.000Z",
      "modified": "2026-09-15T21:00:00.000Z",
      "name": "Ndumiso Mbanjwa (May26 Cybersecurity)",
      "description": "SOC analyst incident report \u2014 May26 Cybersecurity cohort, SOC Simulator operation Cloud Identity Under Siege.",
      "identity_class": "individual",
      "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
      ]
    },
    {
      "type": "malware",
      "spec_version": "2.1",
      "id": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
      "created": "2025-10-06T08:15:05.000Z",
      "modified": "2026-09-15T21:00:00.000Z",
      "name": "goblob.exe",
      "description": "Unsigned binary run from C:\\\\Users\\marcus.aurelius\\Downloads"
```

that resolved finance-backups.blob.core.windows.net via 217.44.76.238; firewall verdict Allowed.",

```
    "malware_types": [
      "unknown"
    ],
    "is_family": false,
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
  },
  {
    "type": "tool",
    "spec_version": "2.1",
    "id": "tool--7b859f95-19be-56d3-ab7b-43ce60984138",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "AADInternals",
    "description": "PowerShell toolkit; Get-AADIntAccessToken used on CORP-WKS-8821 to obtain a Microsoft Graph access token.",
    "tool_types": [
      "credential-exploitation"
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
  },
  {
    "type": "tool",
    "spec_version": "2.1",
    "id": "tool--73dc6698-1b0b-5d91-ac93-b706a87661f6",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "AzureHound",
    "description": "Azure/Entra ID enumeration tool run as dev_ops_automation, output to /tmp/graph_objects.json.",
    "tool_types": [
      "information-gathering"
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
  },
  {
    "type": "tool",
    "spec_version": "2.1",
    "id": "tool--cc4247e7-9c9a-584c-be50-f16d696b3961",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "kubect1",
    "description": "Kubernetes CLI used to list secrets across all namespaces into /tmp/etcd_secrets.txt and to reach the API server 10.0.2.143:6443.",
    "tool_types": [
      "information-gathering"
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
```

```

        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
},
{
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--596a6c6e-a3e5-50fd-a613-ffa2c4173bfa",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Valid Accounts: Cloud Accounts",
    "description": "Use of marcus.aurelius, dev_ops_automation and
admin.breakglass cloud identities.",
    "external_references": [
        {
            "source_name": "mitre-attack",
            "url": "https://attack.mitre.org/techniques/T1078/004/",
            "external_id": "T1078.004"
        }
    ],
    "kill_chain_phases": [
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "initial-access"
        },
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "persistence"
        },
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "privilege-escalation"
        },
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "stealth"
        }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
},
{
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--b05ba812-51fe-5ca5-9fb6-a9880b1e2795",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Account Manipulation: Additional Cloud Roles",
    "description": "AddAppRoleAssignment by marcus.aurelius from 10.0.1.134.",
    "external_references": [
        {
            "source_name": "mitre-attack",
            "url": "https://attack.mitre.org/techniques/T1098/003/",
            "external_id": "T1098.003"
        }
    ],
    "kill_chain_phases": [
        {

```

```

        "kill_chain_name": "mitre-attack",
        "phase_name": "persistence"
    },
    {
        "kill_chain_name": "mitre-attack",
        "phase_name": "privilege-escalation"
    }
],
"created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
"object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
]
},
{
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--18015a1c-3f46-5b42-aa5b-f519b1453952",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Hide Artifacts: Email Hiding Rules",
    "description": "Listed by the platform's ATT&CK Cloud panel for this
incident.",
    "external_references": [
        {
            "source_name": "mitre-attack",
            "url": "https://attack.mitre.org/techniques/T1564/008/",
            "external_id": "T1564.008"
        }
    ],
    "kill_chain_phases": [
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "stealth"
        }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
},
{
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--8ed1a50a-e707-5ad7-9d81-a24038b93238",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Steal Application Access Token",
    "description": "Get-AADIntAccessToken for Microsoft Graph on CORP-WKS-8821.",
    "external_references": [
        {
            "source_name": "mitre-attack",
            "url": "https://attack.mitre.org/techniques/T1528/",
            "external_id": "T1528"
        }
    ],
    "kill_chain_phases": [
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "credential-access"
        }
    ]
}

```

```

    }
  ],
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
  ]
},
{
  "type": "attack-pattern",
  "spec_version": "2.1",
  "id": "attack-pattern--ac278439-7ee3-5f0f-8ca1-316dbd7e3f9b",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",
  "name": "Unsecured Credentials: Credentials In Files",
  "description": ".env copied from srv-app-prod-01 over HTTP.",
  "external_references": [
    {
      "source_name": "mitre-attack",
      "url": "https://attack.mitre.org/techniques/T1552/001/",
      "external_id": "T1552.001"
    }
  ],
  "kill_chain_phases": [
    {
      "kill_chain_name": "mitre-attack",
      "phase_name": "credential-access"
    }
  ],
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
  ]
},
{
  "type": "attack-pattern",
  "spec_version": "2.1",
  "id": "attack-pattern--5cbf5730-f197-55c8-a6d1-380c01eee327",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",
  "name": "Unsecured Credentials: Container API",
  "description": "kubectrl listed secrets across all namespaces into /tmp/etcd_secrets.txt.",
  "external_references": [
    {
      "source_name": "mitre-attack",
      "url": "https://attack.mitre.org/techniques/T1552/007/",
      "external_id": "T1552.007"
    }
  ],
  "kill_chain_phases": [
    {
      "kill_chain_name": "mitre-attack",
      "phase_name": "credential-access"
    }
  ],
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
  ]
}

```

```

    },
    {
      "type": "attack-pattern",
      "spec_version": "2.1",
      "id": "attack-pattern--f2529058-3fc0-5ac1-9e01-82f65c6110a3",
      "created": "2025-10-06T08:15:05.000Z",
      "modified": "2026-09-15T21:00:00.000Z",
      "name": "Account Discovery: Cloud Account",
      "description": "AzureHound Graph directory enumeration to
/tmp/graph_objects.json.",
      "external_references": [
        {
          "source_name": "mitre-attack",
          "url": "https://attack.mitre.org/techniques/T1087/004/",
          "external_id": "T1087.004"
        }
      ],
      "kill_chain_phases": [
        {
          "kill_chain_name": "mitre-attack",
          "phase_name": "discovery"
        }
      ],
      "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
      "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
      ]
    },
    {
      "type": "attack-pattern",
      "spec_version": "2.1",
      "id": "attack-pattern--250b19ab-cece-58d3-a332-6eb1cc2b9165",
      "created": "2025-10-06T08:15:05.000Z",
      "modified": "2026-09-15T21:00:00.000Z",
      "name": "Cloud Storage Object Discovery",
      "description": "goblob.exe targeting finance-backups.blob.core.windows.net.",
      "external_references": [
        {
          "source_name": "mitre-attack",
          "url": "https://attack.mitre.org/techniques/T1619/",
          "external_id": "T1619"
        }
      ],
      "kill_chain_phases": [
        {
          "kill_chain_name": "mitre-attack",
          "phase_name": "discovery"
        }
      ],
      "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
      "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
      ]
    },
    {
      "type": "attack-pattern",
      "spec_version": "2.1",
      "id": "attack-pattern--3a827a23-e4ba-54d6-945c-fe71d341eb4c",
      "created": "2025-10-06T08:15:05.000Z",

```

```

    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Remote Services",
    "description": "Inbound NSG flow from 10.0.1.134 to srv-app-prod-01.",
    "external_references": [
      {
        "source_name": "mitre-attack",
        "url": "https://attack.mitre.org/techniques/T1021/",
        "external_id": "T1021"
      }
    ],
    "kill_chain_phases": [
      {
        "kill_chain_name": "mitre-attack",
        "phase_name": "lateral-movement"
      }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--5108d341-ac3e-5f85-95af-6ee036eef0da",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Use Alternate Authentication Material: Application Access Token",
    "description": "Graph sign-in from 185.220.101.35 with no MFA claim.",
    "external_references": [
      {
        "source_name": "mitre-attack",
        "url": "https://attack.mitre.org/techniques/T1550/001/",
        "external_id": "T1550.001"
      }
    ],
    "kill_chain_phases": [
      {
        "kill_chain_name": "mitre-attack",
        "phase_name": "lateral-movement"
      }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--4410522a-5281-5e8e-ab7a-227b507a8eed",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Data from Cloud Storage",
    "description": "Objects read from the finance-backups container.",
    "external_references": [
      {
        "source_name": "mitre-attack",
        "url": "https://attack.mitre.org/techniques/T1530/",

```

```

        "external_id": "T1530"
    },
    ],
    "kill_chain_phases": [
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "collection"
        }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
},
{
    "type": "attack-pattern",
    "spec_version": "2.1",
    "id": "attack-pattern--9ddc9503-68a0-5e7e-aa82-f462744846d0",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Transfer Data to Cloud Account",
    "description": "95,000 objects (4.2 GB) read by dev_ops_automation from
185.220.101.35.",
    "external_references": [
        {
            "source_name": "mitre-attack",
            "url": "https://attack.mitre.org/techniques/T1537/",
            "external_id": "T1537"
        }
    ],
    "kill_chain_phases": [
        {
            "kill_chain_name": "mitre-attack",
            "phase_name": "exfiltration"
        }
    ],
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
},
{
    "type": "indicator",
    "spec_version": "2.1",
    "id": "indicator--616474da-822b-5232-8621-481d1e8dd845",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Tor exit used for token replay and storage read",
    "description": "Graph sign-in without MFA 2025-10-08T02:09:53Z; 4.2 GB
finance-backups read 2025-10-14T04:59:52Z. VirusTotal 17/89 (2026-09-15).",
    "indicator_types": [
        "malicious-activity"
    ],
    "pattern": "[ipv4-addr:value = '185.220.101.35']",
    "pattern_type": "stix",
    "valid_from": "2025-10-06T08:15:05.000Z",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
}

```

```

    ]
  },
  {
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--43a82259-4b71-5841-86e3-438fca64ae94",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "indicates",
    "source_ref": "indicator--616474da-822b-5232-8621-481d1e8dd845",
    "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "indicator",
    "spec_version": "2.1",
    "id": "indicator--9a2b8ace-df64-5980-9fcc-2e26a9e70d3d",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Suspicious external endpoint",
    "description": "DNS by goblob.exe and repeated HTTPS by powershell.exe from
10.0.1.134; all Allowed. VirusTotal 0/89 (2026-09-15).",
    "indicator_types": [
      "malicious-activity"
    ],
    "pattern": "[ipv4-addr:value = '217.44.76.238']",
    "pattern_type": "stix",
    "valid_from": "2025-10-06T08:15:05.000Z",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--da04b47e-dc54-5724-93e6-acab36550e46",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "indicates",
    "source_ref": "indicator--9a2b8ace-df64-5980-9fcc-2e26a9e70d3d",
    "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "indicator",
    "spec_version": "2.1",
    "id": "indicator--lea926e9-abc8-5ce9-9d75-7eacf60ae389",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "Internal source of anomalous inbound flow",
    "description": "NSG flow to srv-app-prod-01 at 2025-10-08T22:02:53Z (CORP-WKS-
8821).",

```

```

        "indicator_types": [
            "malicious-activity"
        ],
        "pattern": "[ipv4-addr:value = '10.0.1.134']",
        "pattern_type": "stix",
        "valid_from": "2025-10-06T08:15:05.000Z",
        "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
        "object_marking_refs": [
            "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
        ]
    },
    {
        "type": "relationship",
        "spec_version": "2.1",
        "id": "relationship--376010f5-b745-526b-8643-67f5abf71b35",
        "created": "2025-10-06T08:15:05.000Z",
        "modified": "2026-09-15T21:00:00.000Z",
        "relationship_type": "indicates",
        "source_ref": "indicator--1ea926e9-abc8-5ce9-9d75-7eac60ae389",
        "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
        "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
        "object_marking_refs": [
            "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
        ]
    },
    {
        "type": "indicator",
        "spec_version": "2.1",
        "id": "indicator--9bab1db8-d083-5050-9256-52010a8d6bd7",
        "created": "2025-10-06T08:15:05.000Z",
        "modified": "2026-09-15T21:00:00.000Z",
        "name": "Internal origin of break-glass sign-in",
        "description": "admin.breakglass sign-in without MFA at 2025-10-09T10:59:53Z.",
        "indicator_types": [
            "malicious-activity"
        ],
        "pattern": "[ipv4-addr:value = '10.0.2.69']",
        "pattern_type": "stix",
        "valid_from": "2025-10-06T08:15:05.000Z",
        "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
        "object_marking_refs": [
            "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
        ]
    },
    {
        "type": "relationship",
        "spec_version": "2.1",
        "id": "relationship--b071127f-0893-56b6-9015-202cef448169",
        "created": "2025-10-06T08:15:05.000Z",
        "modified": "2026-09-15T21:00:00.000Z",
        "relationship_type": "indicates",
        "source_ref": "indicator--9bab1db8-d083-5050-9256-52010a8d6bd7",
        "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
        "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
        "object_marking_refs": [
            "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
        ]
    }
],

```

```

{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--16d45145-a92d-586a-baa7-ecea332fffd4",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",
  "name": "Staged directory dump",
  "description": "AzureHound output at /tmp/graph_objects.json.",
  "indicator_types": [
    "malicious-activity"
  ],
  "pattern": "[file:name = 'graph_objects.json']",
  "pattern_type": "stix",
  "valid_from": "2025-10-06T08:15:05.000Z",
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
  ]
},
{
  "type": "relationship",
  "spec_version": "2.1",
  "id": "relationship--7a45f90f-7ef6-5c10-88b7-be66400f0a8e",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",
  "relationship_type": "indicates",
  "source_ref": "indicator--16d45145-a92d-586a-baa7-ecea332fffd4",
  "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
  ]
},
{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--c518119b-b511-5daa-830d-9e75d593481f",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",
  "name": "Staged Kubernetes secrets",
  "description": "kubectrl get secrets --all-namespaces -o yaml output at /tmp/etcd_secrets.txt.",
  "indicator_types": [
    "malicious-activity"
  ],
  "pattern": "[file:name = 'etcd_secrets.txt']",
  "pattern_type": "stix",
  "valid_from": "2025-10-06T08:15:05.000Z",
  "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
  "object_marking_refs": [
    "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
  ]
},
{
  "type": "relationship",
  "spec_version": "2.1",
  "id": "relationship--482f50b0-2375-550a-8209-84b3792f7b17",
  "created": "2025-10-06T08:15:05.000Z",
  "modified": "2026-09-15T21:00:00.000Z",

```

```

    "relationship_type": "indicates",
    "source_ref": "indicator--c518119b-b511-5daa-830d-9e75d593481f",
    "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "indicator",
    "spec_version": "2.1",
    "id": "indicator--9dd4bd20-2c10-5629-8c78-0a3615ffb6cc",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "name": "goblob.exe hash",
    "description": "Unsigned goblob.exe. Not found on VirusTotal (2026-09-15).",
    "indicator_types": [
        "malicious-activity"
    ],
    "pattern": "[file:hashes.'SHA-256' = '7b1f3c9a2e8d4f60a15c83be6d9072f4ab3e1c5d8f902a47b6c1de03f8a25e91']",
    "pattern_type": "stix",
    "valid_from": "2025-10-06T08:15:05.000Z",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--b6e53991-f311-587d-9e12-54d110ea276d",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "indicates",
    "source_ref": "indicator--9dd4bd20-2c10-5629-8c78-0a3615ffb6cc",
    "target_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--c219787c-d7ad-5695-904f-4a85be97d0b9",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--596a6c6e-a3e5-50fd-a613-ffa2c4173bfa",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",

```

```

    "id": "relationship--376a721d-c7d2-5d83-b0e0-38e5ec251f15",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--b05ba812-51fe-5ca5-9fb6-a9880b1e2795",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--51f8b017-bf42-5157-97a9-cdbfcb453cc4",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--18015a1c-3f46-5b42-aa5b-f519b1453952",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--892ed792-9344-5d9b-8489-357fdb350ce5",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--8ed1a50a-e707-5ad7-9d81-a24038b93238",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--7338b82d-5d04-5001-bfc8-b2b3baadda3f",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--ac278439-7ee3-5f0f-8ca1-316dbd7e3f9b",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--cafc59d9-a5c8-5c7c-bdc0-552285e32593",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",

```

```

    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--5cbf5730-f197-55c8-a6d1-380c01eee327",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--3bfbb7e2-2e8b-5301-ac45-7f1d4be017f9",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--f2529058-3fc0-5ac1-9e01-82f65c6110a3",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--f076ea10-8ff5-5c53-856d-1096a3f09f29",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--250b19ab-cece-58d3-a332-6eb1cc2b9165",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--ad85c3d5-58ce-5354-ae71-caf8b0938d4f",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--3a827a23-e4ba-54d6-945c-fe71d341eb4c",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
        "marking-definition--94868c89-83c2-464b-929b-ala8aa3c8487"
    ]
},
{
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--d1ea77ae-d9d6-5f96-a47d-cd03679bf7e7",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--5108d341-ac3e-5f85-95af-6ee036eef0da",

```

```

    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--d91af71f-881c-5218-b451-48699947c9ff",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--4410522a-5281-5e8e-ab7a-227b507a8eed",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  },
  {
    "type": "relationship",
    "spec_version": "2.1",
    "id": "relationship--a54dce4a-a2a2-5bcd-ae08-7c8d29244cb7",
    "created": "2025-10-06T08:15:05.000Z",
    "modified": "2026-09-15T21:00:00.000Z",
    "relationship_type": "uses",
    "source_ref": "malware--3f40c0ee-c361-5842-b870-0b7fc0585ea2",
    "target_ref": "attack-pattern--9ddc9503-68a0-5e7e-aa82-f462744846d0",
    "created_by_ref": "identity--fdc72391-0a7a-5e02-b9d8-1fc5d3d67c55",
    "object_marking_refs": [
      "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
    ]
  }
]
}

```

8. MITRE ATT&CK Mapping

Tactic	Technique	ID
Initial Access, Persistence, Privilege Escalation, Stealth	Valid Accounts: Cloud Accounts	T1078.004
Persistence, Privilege Escalation	Account Manipulation: Additional Cloud Roles	T1098.003
Stealth	Hide Artifacts: Email Hiding Rules	T1564.008
Credential Access	Steal Application Access Token	T1528
Credential Access	Unsecured Credentials: Credentials In Files	T1552.001
Credential Access	Unsecured Credentials: Container API	T1552.007
Discovery	Account Discovery: Cloud Account	T1087.004
Discovery	Cloud Storage Object Discovery	T1619
Lateral Movement	Remote Services	T1021

Tactic	Technique	ID
Lateral Movement	Use Alternate Authentication Material: Application Access Token	T1550.001
Collection	Data from Cloud Storage	T1530
Exfiltration	Transfer Data to Cloud Account	T1537

Eleven of these techniques are taken directly from the platform's "MITRE ATT&CK Cloud — techniques used in this incident" panel (Figs. 11–12), with the tactic placement shown there; T1552.007 comes from the platform's XDR Detected Behaviors panel for the confirmed etcd_secrets.txt activity. T1564.008 is listed because the platform maps it to this incident, although no email-rule record was examined in the scored tasks.

9. Impact & Containment

Impact: the attacker obtained valid cloud access tokens, a full map of the Azure directory, every secret stored in the Kubernetes cluster and the production application's .env configuration, used the emergency break-glass administrator account without MFA, and read 4.2 GB of finance backup data out through Tor. In practical terms, any password, key or token held in the cluster, the application configuration or the directory should be treated as known to the attacker. The firewall allowed every outbound connection involved, so nothing at the perimeter limited the damage. Containment steps, in priority order:

1. Revoke all sessions and refresh tokens for marcus.aurelius, dev_ops_automation and admin.breakglass; reset their credentials and remove the app-role assignment added on 2025-10-10.
2. Rotate every Kubernetes secret, the credentials in srv-app-prod-01's .env file, and the finance-backups storage account keys/SAS tokens, since all of these must be assumed stolen.
3. Isolate CORP-WKS-8821 and investigate the host at 10.0.2.69; remove goblob.exe, AzureHound.exe, /tmp/graph_objects.json and /tmp/etcd_secrets.txt after evidence is preserved.
4. Block 185.220.101[.]35 and 217.44.76[.]238 and deny Tor exit nodes at the firewall and in Conditional Access; replace the Allow-Outbound-DNS-Any and Allow-Outbound-HTTPS-Any rules with restrictive egress rules.
5. Stop srv-app-prod-01 serving files over plain HTTP and review who can reach it directly; confirm with the data owners what finance data the 95,000 objects contained for breach assessment.

10. Conclusion & Recommendations

Final verdict: True Positive. What looked like scattered oddities across nine days form one intrusion: token theft from a user workstation, directory reconnaissance, misuse of a break-glass administrator, harvesting of Kubernetes and application secrets, and a 4.2 GB read of finance backups through Tor. Every finding is backed by the platform's records and its own ATT&CK Cloud mapping. The most

operationally important point is the firewall's ALLOWED verdict: prevention failed completely, and the organisation was protected only by the ability to detect and investigate after the fact.

Recommendations:

1. Implement default-deny egress filtering (including DNS) and block known anonymisation networks such as Tor, so unknown binaries cannot freely reach the internet.
2. Protect break-glass accounts with phishing-resistant MFA, alert on every use, and restrict them to specific admin workstations.
3. Apply Conditional Access and token protection to service accounts such as dev_ops_automation, and alert on Graph token requests from tools like AADInternals and AzureHound.
4. Store Kubernetes and application secrets in a managed vault (such as Azure Key Vault), restrict "list secrets" permissions, and never serve .env files from web servers.
5. Alert on high-volume reads from sensitive storage containers and on unsigned executables launched from user Downloads folders.

11. Evidence / Screenshots

Figure	Caption	Referenced in
Figure 1	The operation header showing the completed 100% badge, the scenario's title, difficulty (Intermediate), SIEM/XDR/FW/CLOUD tool surfaces and points, with the 8/8 task panel	Section 3
Figure 2	The SIEM logon search with the Task 1 Confirmed finding 4624: “[4624] Successful logon for marcus.aurelius” on corp-wks-8821 at 08:59:53 and 14:02:11, plus later sign-ins including admin.breakglass	Task 1
Figure 3	The CLOUD panel NSG flow records with the Task 2 Confirmed finding 10.0.1.134: the only Suspicious flow, 2025-10-08 22:02:53 from 10.0.1.134 to host srv-app-prod-01, result Success	Task 2
Figure 4	The CLOUD panel sign-in record with the Task 3 Confirmed finding 10.0.2.69: admin.breakglass signed in from 10.0.2.69, MFA Not used, result Success, application srv-app-prod-01	Task 3
Figure 5	The SIEM record with the Task 4 Confirmed finding kubectrl.exe: “Network connection detected: kubectrl.exe -> 10.0.2.143:6443” on k8s-node-primary-01	Task 4
Figure 6	Process-creation Event 4688 with the Task 5 Confirmed finding /tmp/graph_objects.json: cmd.exe running AzureHound.exe as dev_ops_automation, output redirected to /tmp/graph_objects.json	Task 5
Figure 7	The XDR process tree for CORP-WKS-8821: goblob.exe resolving finance-backups.blob.core.windows.net via 217.44.76.238:53, PowerShell Get-AADIntAccessToken connecting to 185.220.101.35:443, and AzureHound.exe as dev_ops_automation — all Suspicious	Tasks 5, 8
Figure 8	Process-creation Event 4688 with the Task 6 Confirmed finding /tmp/etcd_secrets.txt:	Task 6

Figure	Caption	Referenced in
	"cmd.exe /c kubectl get secrets --all-namespaces -o yaml >" on CORP-WKS-8821	
Figure 9	Process-creation Event 4688 with the Task 7 Confirmed finding .env: "cmd.exe /c curl -s http://srv-app-prod-01/.env -o C:\Users\..." on CORP-WKS-8821	Task 7
Figure 10	The firewall log for 10.0.1.134 with the Task 8 Confirmed finding allowed: goblob.exe DNS to 217.44.76.238:53 Allowed at 2025-10-06 08:15:03, then HTTPS to 185.220.101.35 and 217.44.76.238 Allowed	Task 8
Figure 11	The platform's MITRE ATT&CK Cloud panel (part 1): techniques used in this incident for Initial Access, Persistence, Privilege Escalation, Stealth and Credential Access	Section 8
Figure 12	The platform's MITRE ATT&CK Cloud panel (part 2): Discovery (T1087.004, T1619), Lateral Movement (T1021, T1550.001), Collection (T1530) and Exfiltration (T1537)	Section 8
Figure 13	VirusTotal results for 185.220.101.35: 17 of 89 vendors flag it, tagged tor (AS 60729, DE)	Section 5
Figure 14	VirusTotal results for 217.44.76.238: 0 of 89 vendors (AS 2856, British Telecommunications, GB)	Section 5

Figure 1. The operation header showing the completed 100% badge, the scenario's title, difficulty (Intermediate), SIEM/XDR/FW/CLOUD tool surfaces and points, with the 8/8 task panel.

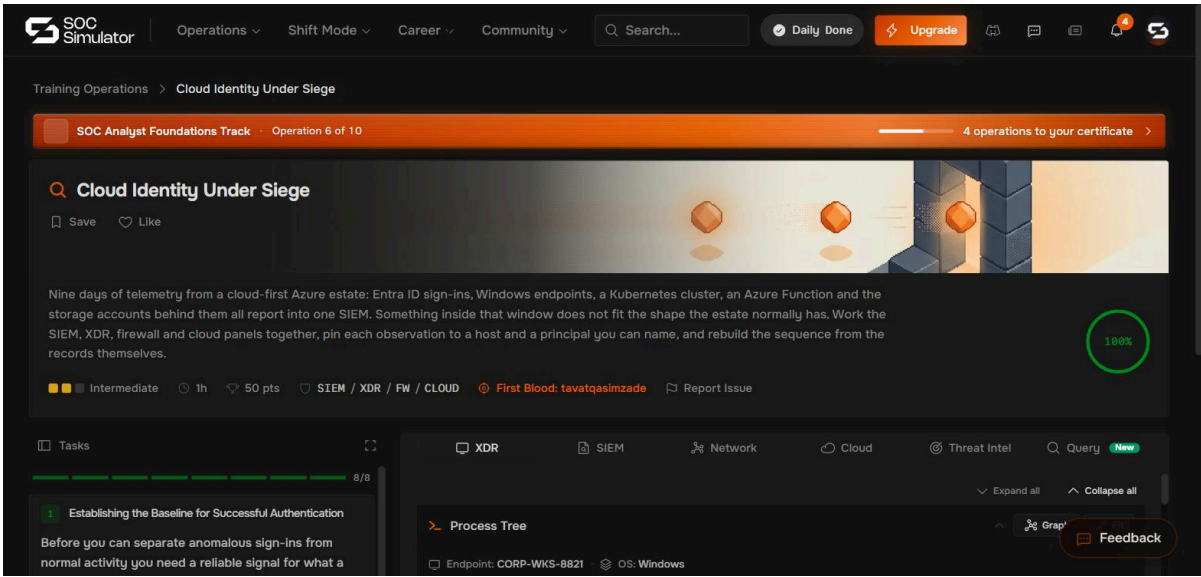


Figure 2. The SIEM logon search with the Task 1 Confirmed finding 4624: “[4624] Successful logon for marcus.aurelius” on corp-wks-8821 at 08:59:53 and 14:02:11, plus later sign-ins including admin.breakglass.

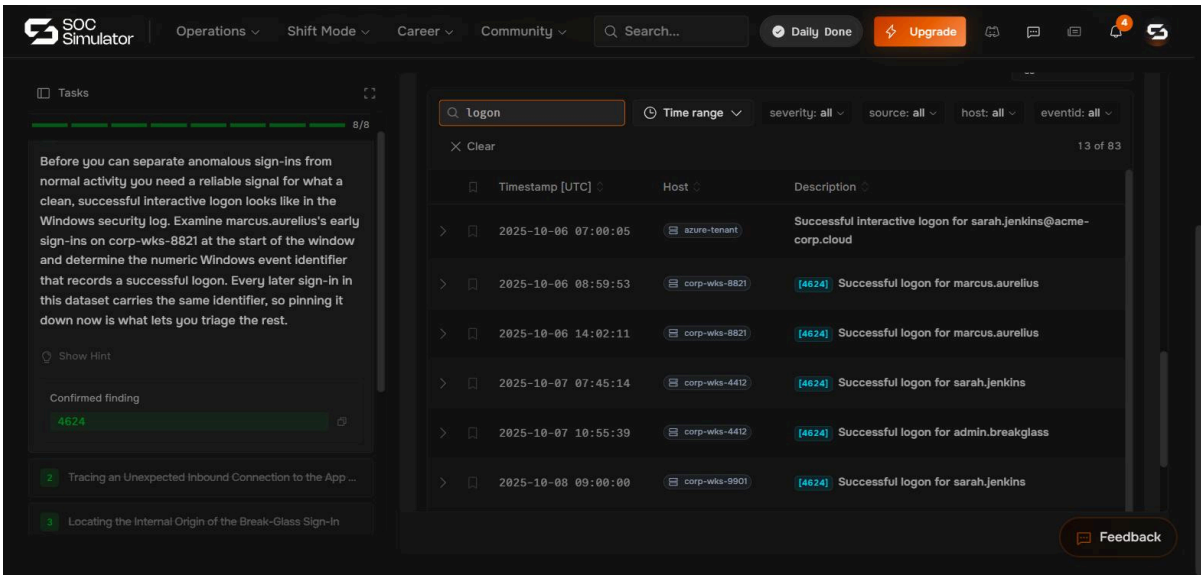


Figure 3. The CLOUD panel NSG flow records with the Task 2 Confirmed finding 10.0.1.134: the only Suspicious flow, 2025-10-08 22:02:53 from 10.0.1.134 to host srv-app-prod-01, result Success.

Cloud Events 53

Search: NetworkFlowRecorded | Time range | severity: all | category: all | Clear | 3 of 53

Timestamp [UTC]	Severity	Event	Category	Principal	Source IP	Suspicious
2025-10-08 08:29:53	High	NetworkFlowRecorded	Network	unknown	10.0.0.6	
2025-10-08 22:02:53	High	NetworkFlowRecorded	Network	unknown	10.0.1.134	Suspicious

Principal Details:

- Name: unknown
- Type: Network

Affected Resources:

- Host: srv-app-prod-01

MITRE ATTACK

Request Details:

- Source: NetworkSecurityGroupFlowLog
- IP: 10.0.1.134
- Result: Success

Feedback

Figure 4. The CLOUD panel sign-in record with the Task 3 Confirmed finding 10.0.2.69: admin.breakglass signed in from 10.0.2.69, MFA Not used, result Success, application srv-app-prod-01.

UserLoggedIn 8

Search: | Time range | type: all | 8 entries

Timestamp [UTC]	Severity	Event	Category	Principal	Source IP	Suspicious
2025-10-08 10:59:53	High	UserLoggedIn	SignInLogs	admin.brea...	10.0.2.69	Suspicious

Principal Details:

- Name: admin.breakglass
- Type: User
- MFA: Not used

Affected Resources:

- Application: srv-app-prod-01

MITRE ATTACK

Request Details:

- Source: AzureActiveDirectory
- IP: 10.0.2.69
- Result: Success

Principals 8

Search: | Time range | type: all | 8 entries

Principal	Type	Events	Failed	Source IPs	First Seen	Last Seen
-----------	------	--------	--------	------------	------------	-----------

Feedback

Figure 5. The SIEM record with the Task 4 Confirmed finding kubecttl.exe: “Network connection detected: kubecttl.exe -> 10.0.2.143:6443” on k8s-node-primary-01.

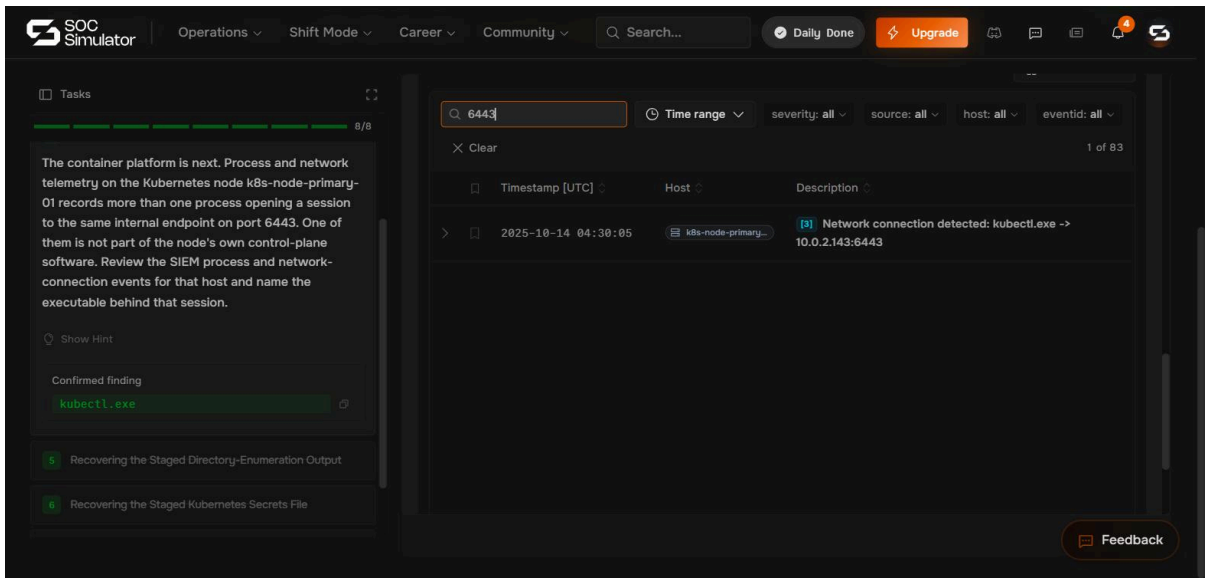


Figure 6. Process-creation Event 4688 with the Task 5 Confirmed finding /tmp/graph_objects.json: cmd.exe running AzureHound.exe as dev_ops_automation, output redirected to /tmp/graph_objects.json.

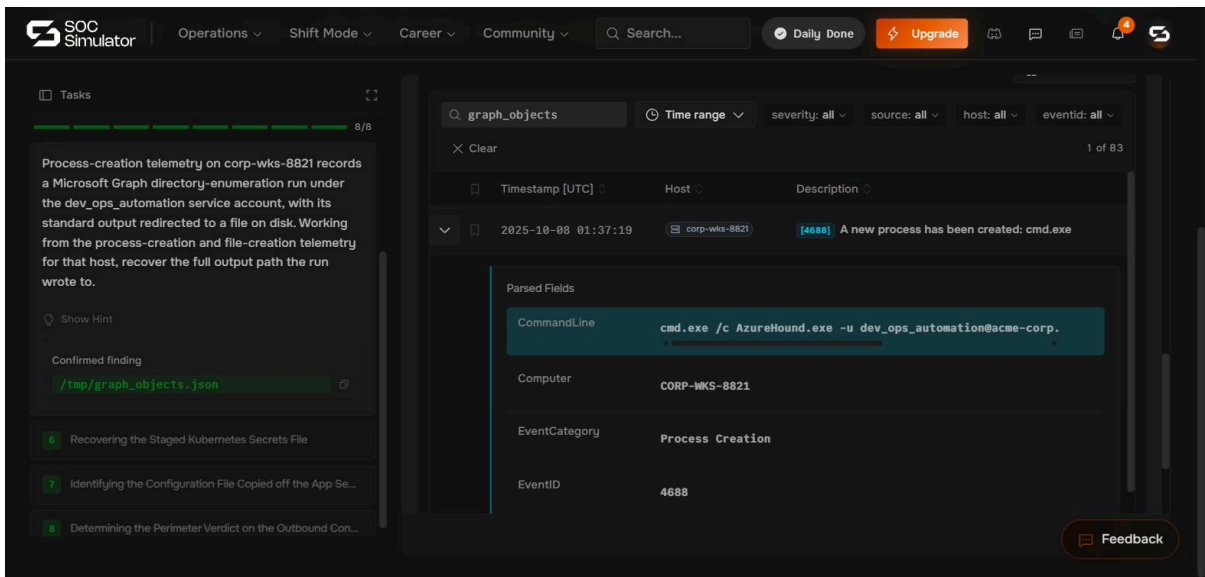


Figure 7. The XDR process tree for CORP-WKS-8821: goblob.exe resolving finance-backups.blob.core.windows.net via 217.44.76.238:53, PowerShell Get-AADIntAccessToken connecting to 185.220.101.35:443, and AzureHound.exe as dev_ops_automation — all Suspicious.

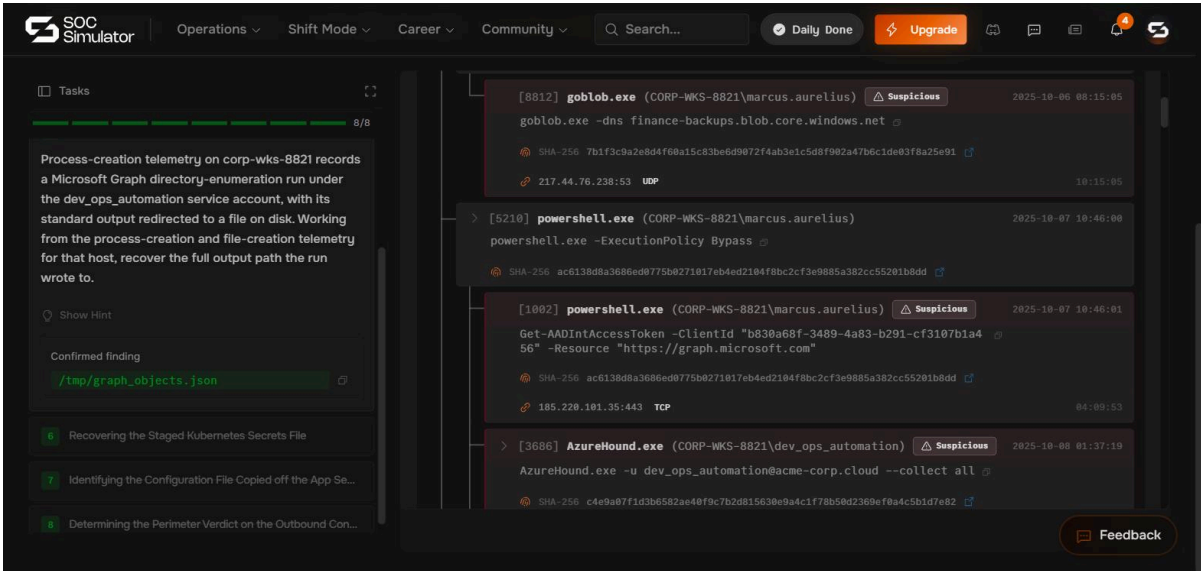


Figure 8. Process-creation Event 4688 with the Task 6 Confirmed finding /tmp/etcd_secrets.txt: "cmd.exe /c kubectl get secrets --all-namespaces -o yaml >" on CORP-WKS-8821.

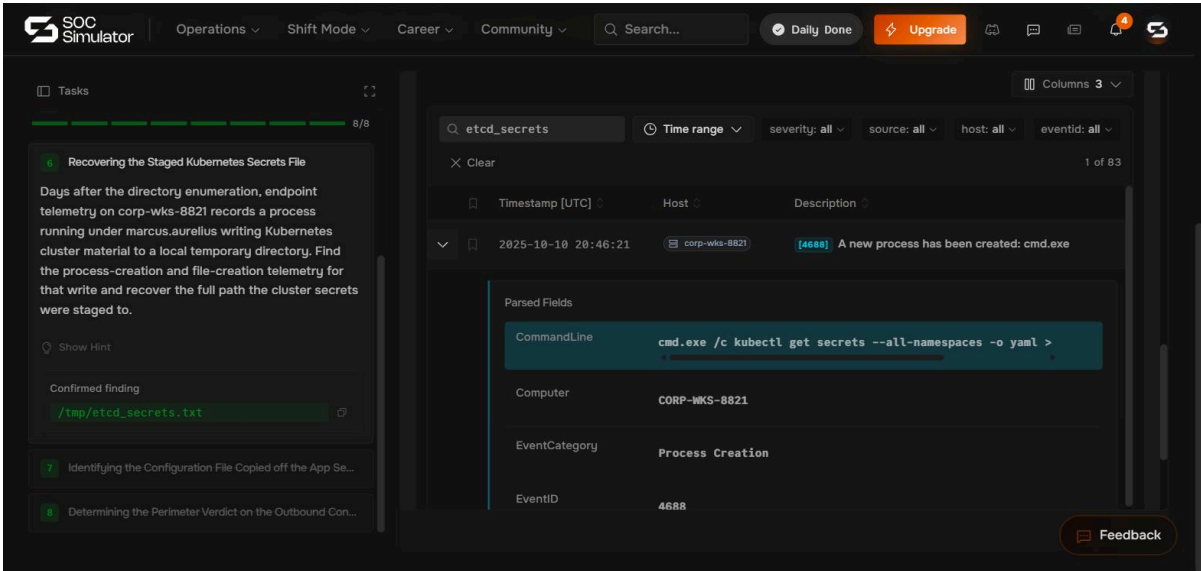


Figure 9. Process-creation Event 4688 with the Task 7 Confirmed finding .env: “cmd.exe /c curl -s http://srv-app-prod-01/.env -o C:\Users\...” on CORP-WKS-8821.

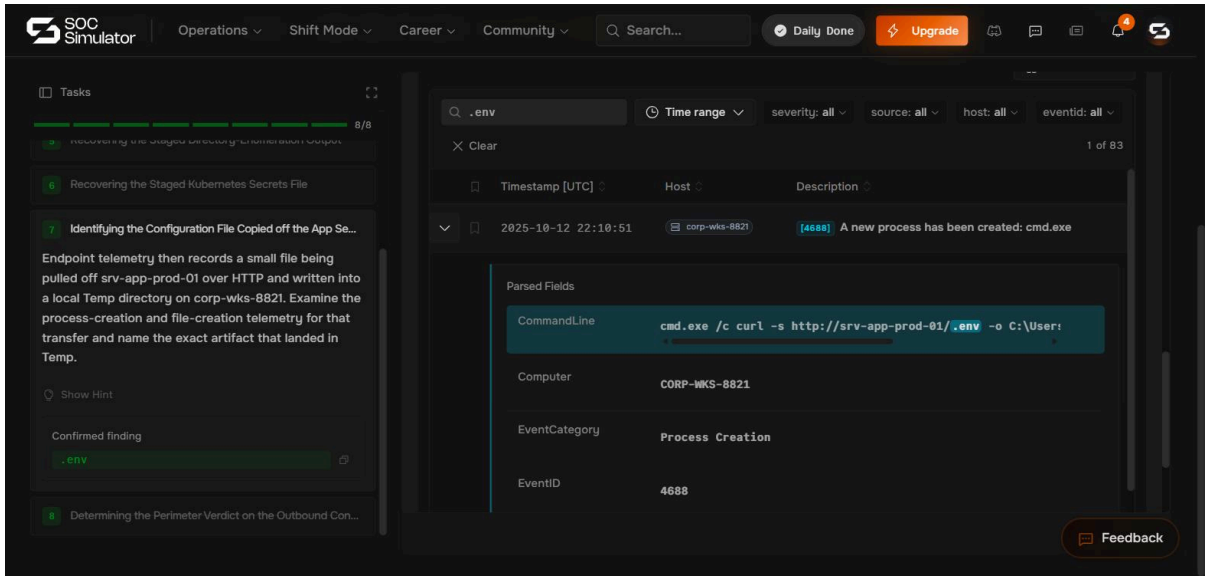


Figure 10. The firewall log for 10.0.1.134 with the Task 8 Confirmed finding allowed: goblob.exe DNS to 217.44.76.238:53 Allowed at 2025-10-06 08:15:03, then HTTPS to 185.220.101.35 and 217.44.76.238 Allowed.

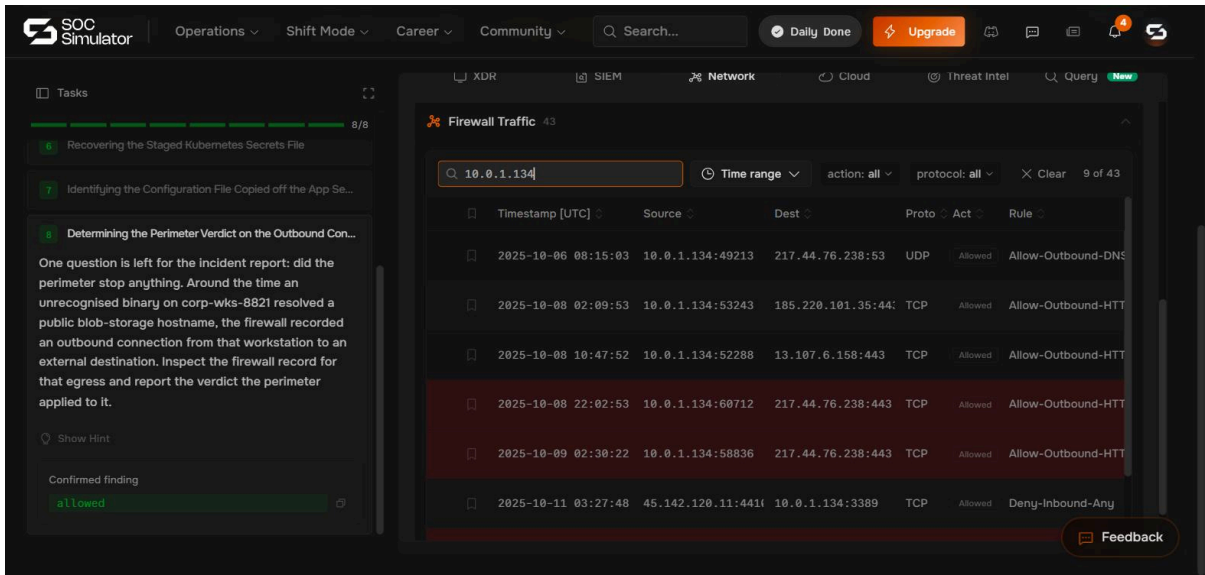


Figure 11. The platform's MITRE ATT&CK Cloud panel (part 1): techniques used in this incident for Initial Access, Persistence, Privilege Escalation, Stealth and Credential Access.

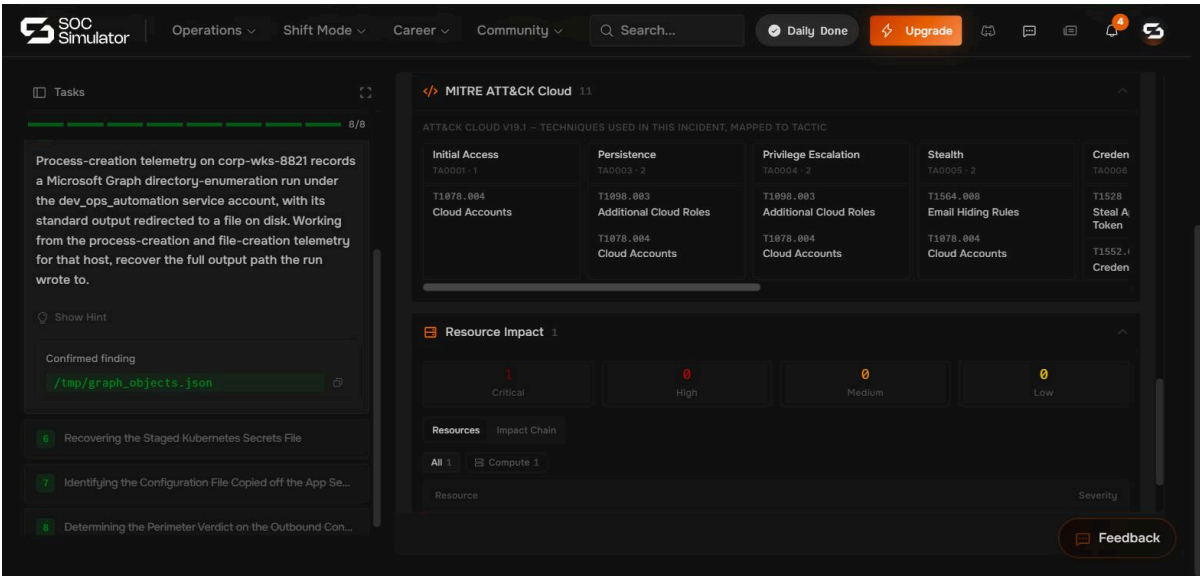


Figure 12. The platform's MITRE ATT&CK Cloud panel (part 2): Discovery (T1087.004, T1619), Lateral Movement (T1021, T1550.001), Collection (T1530) and Exfiltration (T1537).

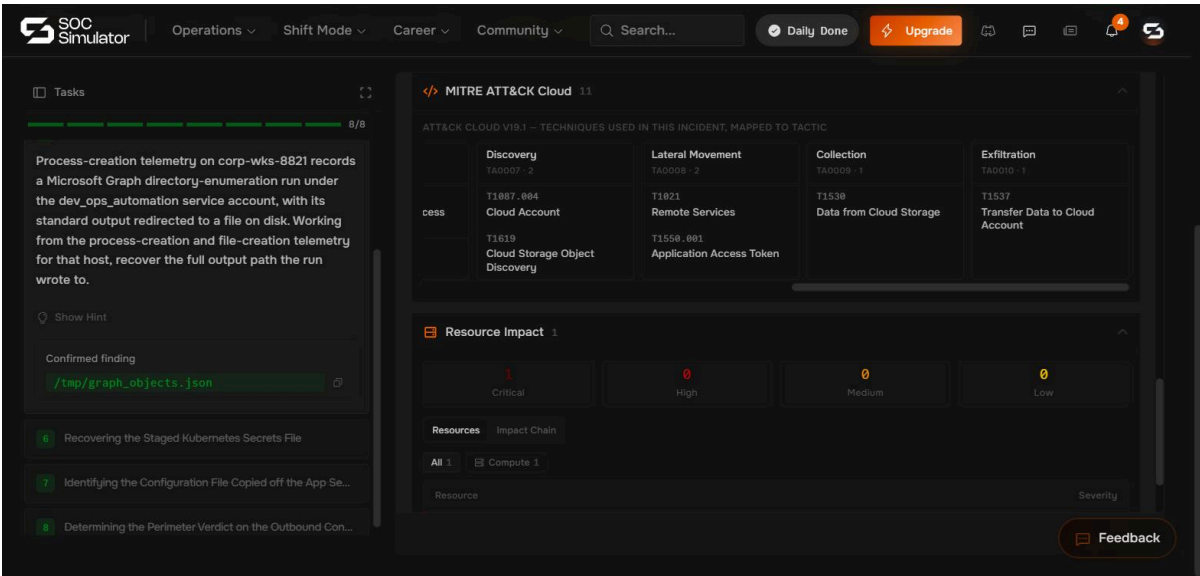


Figure 13. VirusTotal results for 185.220.101.35: 17 of 89 vendors flag it, tagged tor (AS 60729, DE).

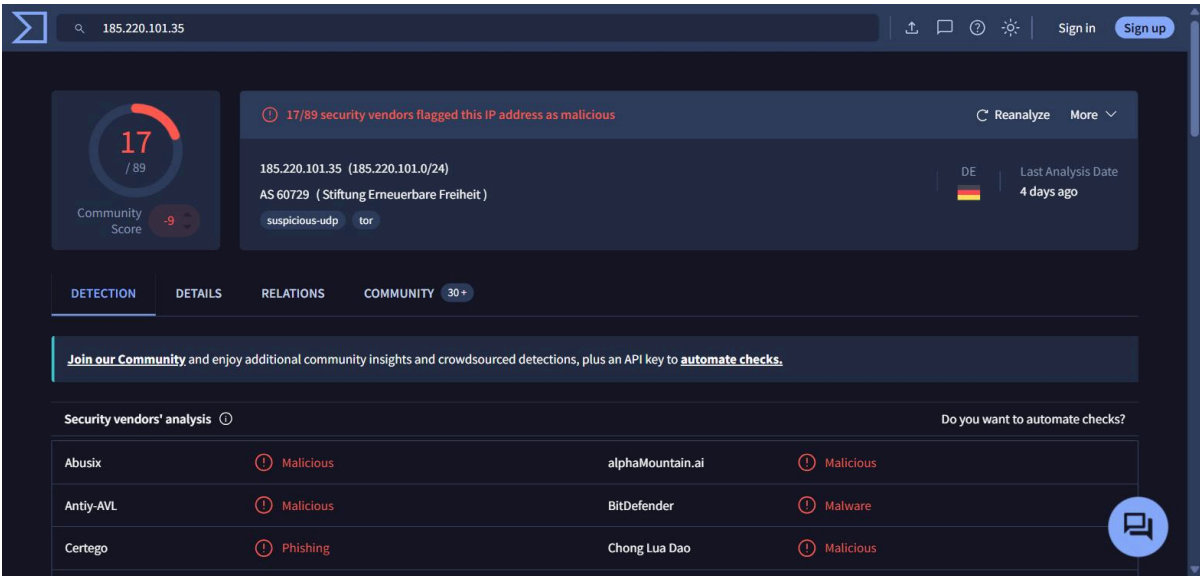


Figure 14. VirusTotal results for 217.44.76.238: 0 of 89 vendors (AS 2856, British Telecommunications, GB).

